

Technology for 4G: LTE or 802.16m?

The term 4G has multiple definitions. In simple terms, it is the much-discussed next-generation network (NGN). It would feature bandwidths as high as 100 Mbps to 1 Gbps, greater convergence of the mobile and the Internet, much more seamless worldwide roaming, and so on. It is hoped that 802.16m will usher in 4G. However, it is not the only 'next generation' technology. There is another on the anvil.

"Both IEEE's 802.16m and the 3rd Generation Partnership Project's Long-Term Evolution Release 10 (3GPP LTE-Release 10) are candidate technologies for implementing 4G. The 802.16m standard will be complete in 2010. 3GPP LTE-Release 10 is still under development and expected to complete in 2011. In terms of performance, 802.16m and LTE Release 10 are similar in capabilities, with 802.16m having a slight edge in terms of performance," sums up Puthenkulam.

Both 802.16m WiMAX and LTE have a similar single-cell geographical coverage subject to the same frequency of operation, simplified flat-packet transport based architectures, latest adaptive communications techniques including OFDM, adaptive modulation to maintain consistent bit-error rate, hybrid automatic request and cyclic prefixing for handling transmission errors, security, QoS, radios and antennae.

But obviously, they are not the same! Dr Borkar helps you understand the differences better. WiMAX is more data-centric for fixed and nomadic applications, whereas LTE is more focused on mobility and wide-area coverage.

The handover capabilities of 802.16e WiMAX are somewhat unproven and the migration path from 802.16e to 802.16m WiMAX is somewhat questionable but is not disruptive. LTE has a legacy of design and deployment experience in the mobility arena since it comes from the mature 3GPP family of technologies. However, a changeover from existing 3G WCDMA/HSPA

Technology behind the 802.16m promise

The improved capacity is derived from the use of multi-carrier operations and high bandwidth (100 MHz compared to 20 MHz for 802.16e). Two or even more carriers, not necessarily in adjacent bands, can be used for transferring data. The physical channels may be of different bandwidths on contiguous or non-contiguous frequency bands; e.g., 5, 10 and 20 MHz. These may use different duplexing modes such as FDD or TDD, or a mix of bidirectional and broadcast-only carriers. 802.16m supports both TDD and FDD for easier and more effective deployments, whereas 802.16e is focused more on TDD. All this enables the capacity to be as high as 1 gigabit per second (Gbps).

802.16m also addresses the inflexibility issue inherent in the 802.16e frame structure. The super frame (20-millisecond long) is segmented into four 5ms frames and further into eight 0.617ms sub-frames, thus providing lower latencies for both control and user planes. Hence faster network access and faster repetition of faulty data blocks are achieved, as the devices have multiple transmission opportunities per frame.

Better handover speeds are achieved by better radio resource management and less handover interrupt time.

The 802.16m base station possesses protocol structure to support multi-radio coexistence for interfacing with different 802.xx standards and other cellular technologies. It possesses enhanced key management at media access control (MAC Layer 2) for better security.

The capabilities of the multi-hop relay station allow 'mesh networks' to be formed. It includes synchronisation capabilities, i.e., the ability to synchronise frame timing and frame counters across the entire system deployed in a given geographic area.

The self-configuration capability of 802.16m automates the configuration of base station parameters to optimise network performance, e.g., Quality of Service (QoS), network efficiency, throughput, frequency reuse, and cell coverage and cell capacity. This results in plug-and-play installation of network nodes and cells based on self-adaptation of the initial configuration, including the update of neighbouring nodes and neighbouring cells as well as the means for fast compensation and reconfiguration in case of failure. Base stations have the capability to adjust their power settings and organise themselves more efficiently with neighbouring base stations by using interference measurement data from different devices.

802.16m is also expected to support additional band profiles of 450-470, 1710-2025, and 2110-2200 MHz.

—Dr Suresh Borkar, a telecom consultant and member of faculty at the Illinois Institute of Technology, Chicago

networks to LTE would be rather disruptive, as LTE includes OFDM.

Plus, WiMAX is designed from the start to rely on Internet Engineering Task Force (IETF) based packet transport protocols and entities, whereas 3G cellular and LTE have still to transition completely towards the IETF-based environment.

3G cellular and LTE have the potential advantages of terminal cost and economy of scale. While big companies create their own chips, etc for 3G cellular and LTE, major WiMAX chip manufacturers rely on smaller start-ups and package their chips, waiting to see whether the momentum will move towards WiMAX.

"Both 802.16m WiMAX and LTE have their place in the 4G framework.

LTE and its overall network is focused on wide-area mobility coverage, whereas WiMAX on metropolitan and rural high-bandwidth data coverage for stationary and nomadic applications. WiMAX is also a prime candidate for use as a backhaul for WiFi, WiMAX and cellular networks," says Dr Borkar.

How fast will the pick-up be?

Let us quickly weigh the situation. 802.16m and LTE are the two technologies competing to rule the 4G era when it dawns. 802.16m is almost ready for standardisation. Equipment will be available by 2011-12. By that time, LTE might also be ready for deployment. Both have their advantages and disad-